

Object Oriented Programming

Phase 1: Requirements and Use Cases

Project: Clinic Appointment System

1. Introduction

The Clinic Appointment System is a Java-based console application designed to manage clinic operations such as adding doctors, registering patients, booking appointments, and viewing records. It uses OOP concepts like classes, objects, and ArrayList.

2. Project Objective

- Add doctor details
- Add patient details
- Book appointments
- View appointments
- Manage records

3. Actor

Admin / Receptionist performs all operations.

4. Use Cases

Add Doctor, Add Patient, Book Appointment, View Appointments, Exit

5. Book Appointment Flow

Enter Doctor ID → Enter Patient ID → Enter Date → Validate → Book Appointment

Program (Java Implementation)

```
import java.util.*;

class Doctor {
    int id;
    String name;
    String specialization;

    Doctor(int id, String name, String specialization) {
        this.id = id;
        this.name = name;
        this.specialization = specialization;
    }
}

class Patient {
```

```

    int id;
    String name;
    int age;

    Patient(int id, String name, int age) {
        this.id = id;
        this.name = name;
        this.age = age;
    }
}

class Appointment {
    Doctor doctor;
    Patient patient;
    String date;

    Appointment(Doctor doctor, Patient patient, String date) {
        this.doctor = doctor;
        this.patient = patient;
        this.date = date;
    }
}

```

6. Class Design

Doctor(id, name, specialization)
 Patient(id, name, age)
 Appointment(doctor, patient, date)
 Main class controls system.

7. Database Design

Doctor(doctor_id, name, specialization)
 Patient(patient_id, name, age)
 Appointment(appointment_id, doctor_id, patient_id, date)

9. Conclusion

The Clinic Appointment System successfully demonstrates the application of Object-Oriented Programming concepts in solving real-world problems. It provides a structured approach to managing doctors, patients, and appointments efficiently. The system ensures simplicity, usability, and organized data handling. Furthermore, it can be enhanced by integrating database connectivity, graphical user interface (GUI), and advanced features such as appointment rescheduling, authentication, and report generation, making it a scalable and practical solution for modern healthcare management.